High-speed and Low-latency 3D Sensing with a Parallel-bus Pattern

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Abstract

High-speed 3D shape sensing is an essential technology for three-dimensional recognition and manipulation in dynamic scenes. However, conventional high-speed sensing methods mainly focus on the image capturing speed and the actual processing time to obtain a point cloud is not optimized in the measurement scheme and sensing pattern configuration. On the other hand, measurement latency is critical to respond in real-time and physically handle the dynamic scenes.

This paper introduces a physically stereo-rectified projector-camera system for high-speed and low-latency 3D sensing with fast sequential memory access in the decoding process. Moreover, we configure a structured light pattern named "Parallel-bus pattern" with a De Bruijn torus and clock lines to maximize information density and robustly decode the pattern as data transfer in a parallel bus interface. We measured dynamically moving and deforming objects with the proposed system and evaluated the measurement performance. As a result, the developed 3D sensing system with the parallel-bus pattern achieved 27 K-points measurement at higher than 1000 fps with 0.336 ms latency and 0.838 mm accuracy on average.

1. Introduction

Nowadays, 3D shape sensing is an essential technology for various fields, including digital archiving and manufacturing. Additionally, demand for high-speed and real-time 3D sensing techniques is growing to speed up the industry mentioned above and to recognize and immediately respond to dynamically moving or deforming objects in the fields of robotics, human-computer interface, and augmented/virtual reality. Furthermore, the sensing speed is also required for the case that a measurement system itself will move. However, some 3D sensing methods are unsuitable for dynamic scenes due to the measurement principles.

The conventional 3D sensing methods are based on the triangulation or time-of-flight (ToF) principle. The direct ToF method is theoretically suitable for high-speed sensing. However, it has difficulty forming a two-dimensional array and high-speed scanning, so indirect ToF methods based on temporally encoded lighting are widely used [11, 1]. Besides, active stereo methods are computationally low cost and robust among triangulation-based methods, and representative methods are also based on temporally encoded structured light such as Gray code and phase shift patterns. However, these methods based on multiple samples or frames are inappropriate for dynamic scenes since inter-frame motion and deformation of a target cause erroneous matching and artifacts in the results. Consequently, the active stereo method with a single structured light pattern will be a potential candidate for 3D sensing in dynamic scenes. Besides, one-shot sensing methods based on a fixed pattern are practically preferable because a specialized device like a high-speed projector is not required.

Hence, some methods using a single spatially encoded pattern have been proposed for high-speed 3D shape sensing. Nevertheless, the conventional high-speed sensing methods mainly focus on the high-speed image capturing, and the decoding speed has low priority [23, 13]. Thereby, the processing time from image input to 3D shape output is not necessarily short in the conventional methods. On the other hand, the demand for high-speed sensing is in low latency rather than a high frame rate because the real-time response is required to recognize and physically handle the dynamic scenes such as dynamic robot manipulation [7, 24], autonomous driving [4, 17], and dynamic projection mapping [20, 6].

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Then, we propose a high-speed and low-latency 3D shape sensing method using a structured light pattern named "Parallel-bus pattern." We focus not only on the high-speed image capturing with a single projection pattern but also on pattern structure for low-latency decoding algorithm considering general computer architecture. Computational measurement is, so to speak, information communication between the real world and a computer, and so some essential techniques in communication protocols and data transfer methods can be introduced. We use an analogy to parallel bus interface and develop the pattern and algorithm, which can exploit sequential memory access and bit operations for high-speed decoding. This paper introduces the mathematical background, a pattern construction method, and practical applications of the parallel-bus pattern. Furthermore, we evaluate the speed, density, and accuracy of the proposed method and demonstrate high-speed measurement of dynamically moving or deforming objects with low latency.

2. Related works

In 3D sensing based on light, the distance information must be embedded in the properties of light; intensity, the position of a waypoint, direction of propagation, wavelength, polarization, and the temporal changes of them. Although a temporally encoded 3D sensing method has achieved 500 Hz high-speed measurement [8, 18], the deterioration due to inter-frame motion and deformation is inevitable [27, 3]. Besides, measurement methods based on the direction or polarization of light require a specialized system component [22, 12], and intensity gradation and wavelength are easily affected by target materials and color demosaicing [28]. Hence, a single binary monochrome spatially encoded pattern has the potential for measurement of dynamic scenes by a general projector-camera system. With a spatially encoded pattern, the reduction of measurement density is inevitable. Nevertheless, surface normal sensing can more accurately obtain dense depth information because of the frequency characteristics [21]. Moreover, simultaneous measurement and integration of depth and normal can be performed in real-time [19]. Therefore, we mainly focus on speeding up depth sensing in this paper but mathematically maximize the measurement density under the sensing strategy.

In binary monochrome spatially encoded patterns, pseudo-noise sequence (PN-sequence) plays a significant role in identifying the phase on a projection image from a captured image. In particular, a De Bruijn sequence, maximum-length sequence (M-sequence), and maximum-length array (M-array) have been utilized since they belong to two-level autocorrelation function sequence or array and realize the maximum efficiency to embed information [10]. However, the general projector-camera layout two-dimensionally distorts the projection pattern and causes an NP-complete problem [15] and random and slow memory access on a camera image to scan and recognize the distorted pattern. Then, our proposal assumes that the projector-camera system is physically stereo-rectified in advance. The assumption needs manual layout adjustment in calibration but enables high-speed sequential memory access to identify the phase. This setup can determine the phase by scanning the single fixed line on the camera image. However, a De Bruijn sequence and M-sequence can be long in the line direction, and so decoding can be often failed at discontinuous surfaces across the line. On the other hand, an M-array can be configured as a shorter length in the line direction but mathematically needs at least three lines. Therefore, in contrast, an M-array can be wide in the direction perpendicular to the scanning line. For this reason, we introduce a De Bruijn torus to the parallel-bus pattern in the following section and realize high-speed decoding and high-density measurement simultaneously.

3. Parallel-bus pattern

3.1. Features

To clarify the advantages and system requirements, we list the features of our 3D sensing method using the parallel-bus pattern as follows.

SYSTEM REQUIREMENTS. (A1) A single pair of a projector and a camera. (A2) Physical stereo rectification in advance. (A3) Stereo calibration for intrinsic/extrinsic parameters.

FEATURES. (B1) A single binary monochrome spatially encoded pattern (robust to motion and material). (B2) One-shot measurement (high speed). (B3) Decoding by sequential memory access and bit operations (low latency). (B4) Configured with a De Bruijn torus (high density).

3.2. De Bruijn torus

To increase the amount of information embedded in a pattern, we introduce a De Bruijn torus. The features of De Bruijn tori have been researched in the field of combinatorial mathematics [5, 9]. However, most of the research mainly focuses on square De Bruijn tori, which are not used in our proposal. Accordingly, this paper briefly introduces features related to the parallel-bus pattern. A De Bruijn torus is the 2D extension of a De Bruijn sequence and one of the most efficient patterns to embed information densely. Every $m \times n$-bit binary submatrix appears exactly once with wraparound, and so the torus can express 2^{mn} discrete phases. The torus can be configured by at least two lines ($m \geqq 2$. De Bruijn sequences stand for $m = 1$.) contrary to M-array and can realize both sequential memory access and compactness. In this paper, we only use De Bruijn tori for a $2 \times n$-bit submatrix as shown in Fig. 1. Be-

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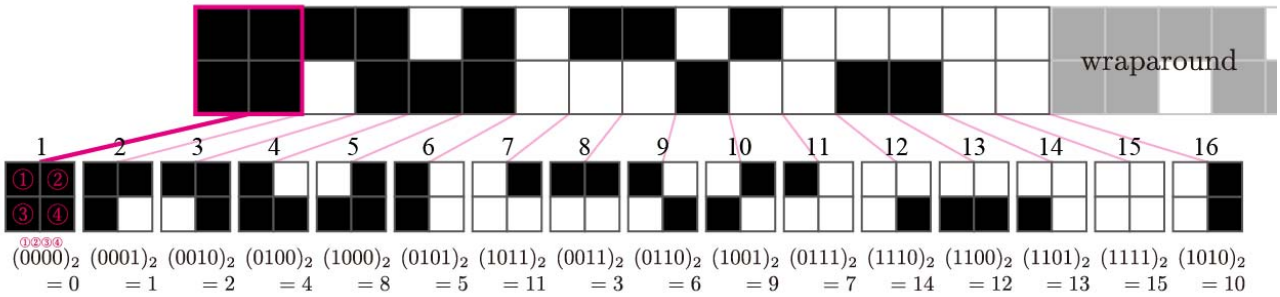


Figure 1. De Bruijn torus. One of De Bruijn tori for a 2×2-bit submatrix is shown as an example. The reading order of bits is arbitrary.

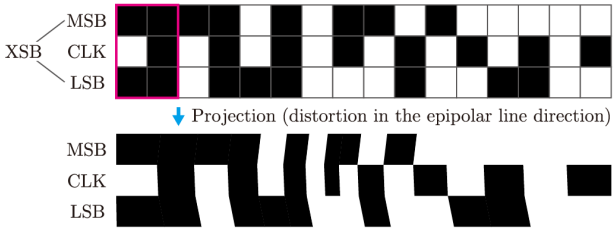


Figure 2. Clock (CLK) and data (XSB) lines. The distortion in this figure assumes perfectly horizontal epipolar lines formed by a physically stereo-rectified setup.

sides, the pattern that swaps the two lines is also a De Bruijn torus.

3.3. Clock and data lines

A De Bruijn torus helps identify the phase. However, the pattern will stretch in an epipolar line direction by projection, and pattern recognition would not be easy for a simple and fast algorithm. In particular, it is challenging to distinguish the same successive numbers in the pattern. The same problem occurs in a parallel bus interface but is solved by various synchronous/asynchronous communication techniques. In this paper, we liken the epipolar lines to parallel buses and introduce a clock line as a data positioner. Figure 2 shows the essential components of the parallel-bus pattern. The proposed pattern has a clock (CLK) line and two data (XSB) lines. The XSB lines consist of the most significant bit (MSB) line and the least significant bit (LSB) line and configure a De Bruijn torus for a $2 \times n$-bit submatrix. By referring to the CLK line, every bit of XSB lines can be easily located, and inefficient data access can be omitted. Note that the clock signal provides triggers to read settled data signals in communication, but CLK and XSB bits simultaneously change in the parallel-bus pattern.

3.4. Decoding

Although a De Bruijn torus includes all possible binary submatrices, they appear in a pseudo-random order. Therefore, a mapping function, which converts a decoded pattern to a phase, is required to obtain the phase. With the mapping function f for a $2 \times n$-bit submatrix, the high-speed decoding algorithm is simply described as Algorithm 1. In the algorithm, $\mathrm{CLK}[\,], \mathrm{MSB}[\,]$, and $\mathrm{LSB}[\,]$ denote W-length pixel arrays in CLK and XSB lines. Thanks to the physically stereo-rectified setup, each array would be on sequential memory space and easily accessed at high speed. Moreover, the algorithm is implementable mainly by bit operations, and so pattern recognition can be immediately performed. In the pseudo program, the fixed threshold I_{th} is used to binarize pixel values for simplification. However, an adaptive threshold is more practical for robustness to the shading and material of a target. Besides, the phase ϕ found at the camera image coordinate u_c is finally converted to a projection image coordinate u_p by another mapping function g. Hence, the composite mapping $g \circ f$ can speed up the calculation without deriving the phase. On the other hand, the discrete phase contains information about surface continuity. For this reason, the phase is worth being calculated when the surface is reconstructed, considering continuity and segmentation.

3.5. Pattern configuration

Configuration of a De Bruijn torus for a $2 \times n$-bit submatrix is a Hamiltonian path problem. On the contrary, the problem is convertible to an Eulerian path problem by a De Bruijn graph, as shown in Fig. 3. Therefore, desired De Bruijn tori are easily obtained as De Bruijn sequences [30].

By combining CLK lines, the parallel-bus pattern is configured, as shown in Fig. 4. In the pattern, a CLK line and an XSB line appear alternately, and an MSB line and an LSB line also appear alternately in the XSB lines. Since the pattern with swapped lines is also a De Bruijn torus, the alternated pattern is also decodable with a corresponding mapping function f'. Moreover, the mapping function f is obviously available by changing the reading order of bits. As a result, the pattern formed by a De Bruijn torus for a $2 \times n$-bit submatrix with N_{CLK} CLK lines has $2^{2n} N_{\mathrm{CLK}}$ identifiable points in a simple configuration. Furthermore,

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Algorithm 1 Decoding algorithm to find a phase ϕ at u_c. The algorithm assumes that True $= 1$ and False $= 0$.

Input: n, W, CLK[ ], MSB[ ], LSB[ ], f[ ], I_{th}
Output: ϕ[ ], u_c[ ]
  {Initialization}
  code $\leftarrow 0$, $i_0 \leftarrow 0$, $j \leftarrow 0$, $P_0 \leftarrow 0$
  {Decoding}
  for $i \leftarrow 0$ to $W-1$ **do**
    $P_1 \leftarrow CLK[i] > I_{\text{th}}$
    if $P_0 \neq P_1$ **then**
      $P_0 \leftarrow P_1$
      $t \leftarrow (i_0 + i) \gg 1$
      $i_0 \leftarrow i$
      code $\ll 1$
      code &=~((1 $\ll n$) + 1)
      code | = ((MSB[t] $> I_{\text{th}}$) $\ll n$) | (LSB[t] $> I_{\text{th}}$)
      $\phi[j] \leftarrow f$[code]
      $u_c[j] \leftarrow i$
      j += 1
    end if
  end for

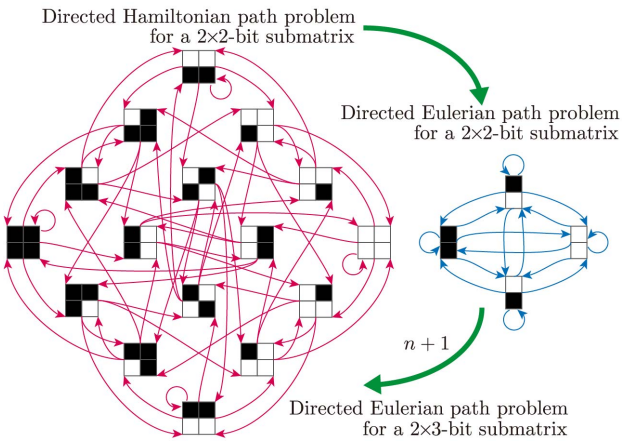


Figure 3. De Bruijn graph for 2×2-bit and 2×3-bit submatrices.

techniques to improve the number of points are described in Sec. 3.6.

Meanwhile, an actual projector-camera system involves lens distortion and misalignment. Although the projection lens distortion can be compensated by correcting the projection image in advance, the camera lens distortion is inevitable. Consequently, the assumption of a physically stereo-rectified setup does not hold perfectly. Then we configure the parallel-bus pattern with variable line width, as shown in Fig. 5. As a result, sequential memory access will be available in a decoding process even with an imperfect alignment. Moreover, the obtained camera image coordinates (u_c, v_c) are numerically corrected according to the lens distortion model instead of correcting the whole image.

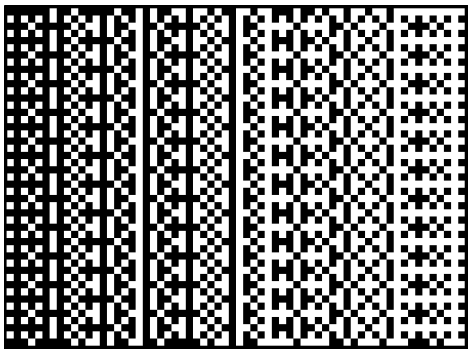


Figure 4. Structure of a parallel-bus pattern. This example is based on a 2×3-bit submatrix and configured by arranging 23 horizontal CLK lines ($n = 3$, $N_{\text{CLK}} = 23$).

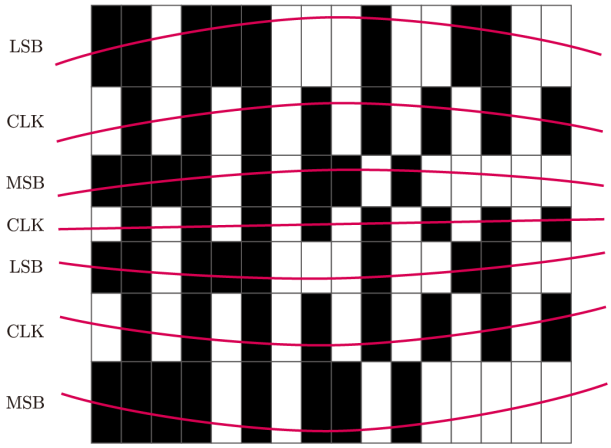


Figure 5. Parallel-bus pattern configured by the variable line width. The lens distortion and misalignment are exaggerated in this figure.

3.6. Odd-bit pattern

In the parallel-bus pattern, the XSB lines are responsible for $2n$-bit information. Furthermore, the CLK line is invertible and can contain 1-bit information additionally. Therefore, the parallel-bus pattern can include $2^{2n+1}N_{\text{CLK}}$ identifiable points in total by stitching an inverted clock pattern, as shown in Fig. 6. There are two ways to stitch the patterns; shrunk CLKs and expanded XSBs. In shrunk CLKs method, the $(n-1)$th and nth CLK bits from the seam are shrunk to half. Thereby, the method requires more than one pixel for each bit. On the other hand, the $(n-1)$th and nth XSB bits from the seam are expanded to 1.5 times in the expanded XSBs method. This method is more robust to be read but lengthens the pattern. Besides, the first code will appear again at the $(n-1)$th submatrix from the seam in both cases. This redundant code can cause erroneous decoding, but the correction is easy because the original first

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code will not appear in the middle of the pattern. Similarly, repeated patterns can configure a projection pattern if phase unwrapping techniques are available.

Meanwhile, the CLK bit changes bit by bit, and so the additional 1-bit information is also helpful for parity check. Furthermore, advanced error correction techniques can be introduced as general data transfer. On the other hand, densification based on inference and interpolation techniques with information such as continuity of surface and adjacency relations is not the focus of this paper.

3.7. Bit inversion

The parallel-bus pattern is a binary monochrome pattern, so the inverted pattern works appropriately. Moreover, the sum of the pattern and the inversion is a uniformly bright image, and exposure or numerical addition of the two frames can work as an acquisition of shading or color while 3D sensing. Simultaneously, the combination with the inversion can perform multi-value sensing. The difference between the two frames shows the occluded region as unchanged pixels and realizes a more robust measurement of complicated surfaces and multiple objects. Although these applications with multiple patterns cause deterioration of the performance in dynamic scenes and require high-speed switching of projection images, binary monochrome patterns can be switched at a much higher speed than 8-bit and 24-bit patterns and are practically effective.

4. Measurement

4.1. System configuration

Figure 7 shows the sensing system composed of a high-speed camera (Optronis CP70-2-M-1000, $1920 \times 1080\,\mathrm{px}@1000\,\mathrm{fps}$, FOV $42\,\mathrm{deg.}$) and a projector (Texas Instruments DLPDLCR471TPEVM, $1920{\times}1080\,\mathrm{px}@60\,\mathrm{fps}$). The camera and projector form a vertical stereo setup, and the camera mount has a five degrees-of-freedom fine adjustment mechanism to realize a physically stereo-rectified projector-camera system. In general, camera pixel readout is carried out row by row, and the captured image is stored in row-major order. Therefore, in order to exploit high-speed sequential memory access in decoding, the camera should be arranged so that the row direction is aligned to the epipolar line direction. In the case of the experimental setup in Fig. 7, the camera must be turned over and lies on the side. On the other hand, we uprightly fixed the camera to maximize the measurement range, an intersection of the field of view and projection. This layout needs to transpose image data for sequential memory access but effectively improves the sensing resolution, especially with a commercially available projector, which upwardly projects images.

In experiments, we configured the parallel-bus pattern for a 2×4-bit submatrix with 162 vertical CLK lines, as shown in Fig. 8. We varied the line width from 5 to $8\,\mathrm{px}$ according to the calibration described in Sec. 4.2 and cropped the pattern to match the projection image format. Besides, we prepared a 3D-printed Stanford Bunny, a textured toy of a dinosaur, a handheld volleyball, and a plaster figure of the Venus as measurement targets, as shown in Fig. 7.

4.2. Calibration

Our setup requires additional adjustment before general calibration of intrinsic, extrinsic, and lens distortion parameters to configure a physically stereo-rectified system. First, we manually aligned the camera and estimated the required line widths according to an exaggerated inclination, distortion, and fluctuation of lines projected by the projector, as shown in Fig. 9. After the adjustment, the deviation of a line was about $0.7\,\mathrm{px}$ at the center and about $3.0\,\mathrm{px}$ at the edge. This adjustment does not require rigorous alignment since the variable line width allows imperfect physical stereo rectification. Subsequently, we listed line numbers of CLK and XSB lines in a camera image and performed the general stereo calibration. In the stereo calibration, the baseline length was about $180\,\mathrm{mm}$, and the largest non-diagonal element in the rotation matrix was about 0.04.

4.3. Measurement results

We measured static objects in Fig. 7 and dynamically moving and deforming hand and shirt at $1000\,\mathrm{fps}$. The distance between the system and the targets was about $550\mathrm{mm}$. Figure 10,11 shows the results. The top row in each figure shows captured images, and relevant lines in the images were input to the decoding function as $\mathrm{CLK}[\,]$,$\mathrm{MSB}[\,]$, and $\mathrm{LSB}[\,]$. Incidentally, W is the height of the images in the vertical stereo setup. As a result of decoding and triangulation, the point clouds in the middle row were obtained in real-time. Moreover, we calculated the surfaces in the bottom row from the point clouds by screened Poisson surface reconstruction [14]. Note that the reconstructed surface far from the original point cloud is trimmed in the figures. The figures show that the plausible shapes of the static and dynamic targets were measured qualitatively. On the other hand, we set the lower limit of an adaptive threshold in the experiment to avoid erroneous code detection, and so the point clouds have missing parts in the dark area. Besides, decoding at discontinuity fails since spatially encoded pattern assumes a continuous surface.

5. Evaluation

5.1. Measurement speed

The proposed decoding method was implemented with C++ and performed on CPUs (Intel, Xeon Silver 4116 @2.10 GHz 2 processors). In the program, decoding and

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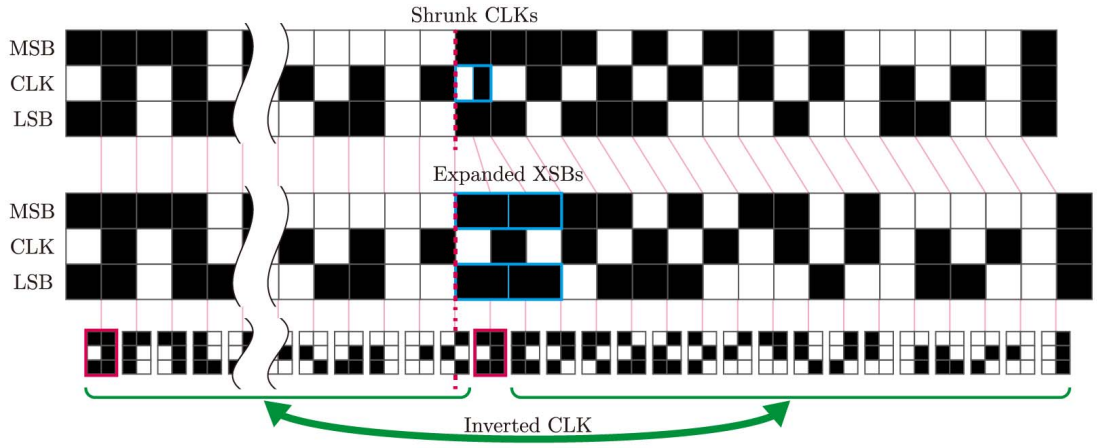


Figure 6. Pattern stitching with shrunk CLKs and expanded XSBs. The shrunk CLKs and expanded XSBs are shown in blue frames. The seam is shown as a red dot line, and redundant codes are shown in red frames.

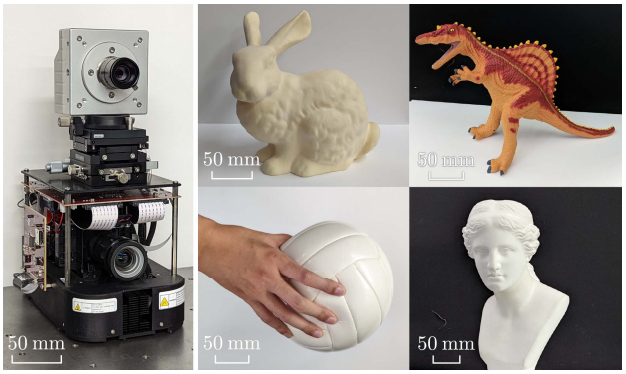


Figure 7. Measurement system and sensing targets.

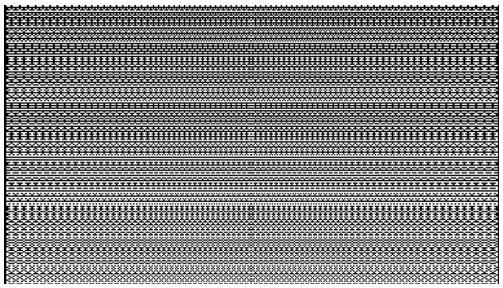


Figure 8. Parallel-bus pattern used in experiments. This pattern is configured with vertical CLK and XSB lines.

triangulation were carried out in parallel with OpenMP. To evaluate measurement speed of the proposed method, we measured a plane and calculated the average processing time for 1000 trials. This plane can cover the whole field of view and cause the greatest calculation amount. As a result, 26713 points were measured from a single frame, and the processing time between image input and point cloud output was $0.336\,\mathrm{ms}$. On the other hand, it was $0.214\,\mathrm{ms}$ when simulating a horizontal stereo setup by transposing the input image in advance. Besides, the processing time between image input and phase output was $0.284\,\mathrm{ms}$ with transpose and $0.141\,\mathrm{ms}$ without transpose. These results yield that the proposed 3D sensing method with the parallel-bus pattern can realize real-time measurement at $1000\,\mathrm{fps}$ in our setup. Moreover, the latency is low enough for up to $3000\,\mathrm{fps}$ real-time measurement with transpose, and $5000\mathrm{fps}$ without transpose.

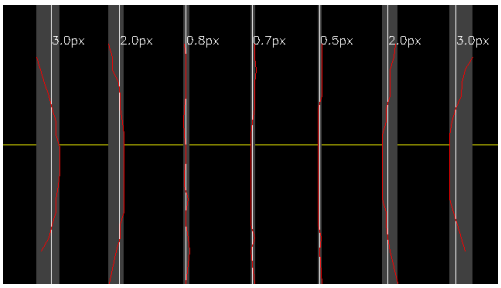


Figure 9. Alignment of a physically stereo-rectified setup. The camera measures projected straight lines. The deviations of the measured lines are exaggerated and shown in red curves. Grey areas and numerical values show the range of deviation. Yellow lines show the center of the image.

5.2. Measurement density

A parallel-bus pattern for a 2×4-bit submatrix with 162 CLK lines can describe up to $2^9 \times 162 = 82944$ identifiable points in a single cycle as mentioned in Sec. 3.6. On the other hand, we used 1-bit information in a CLK line for parity check, and the projection image used in the experiments contained $216 \times 162 = 34992$ identifiable points after cropping the pattern to fit the image size. Furthermore, the actual number of measurement points is limited by the intersection of the field of view and projection. In the measurement of the plane mentioned in Sec. 5.1, the

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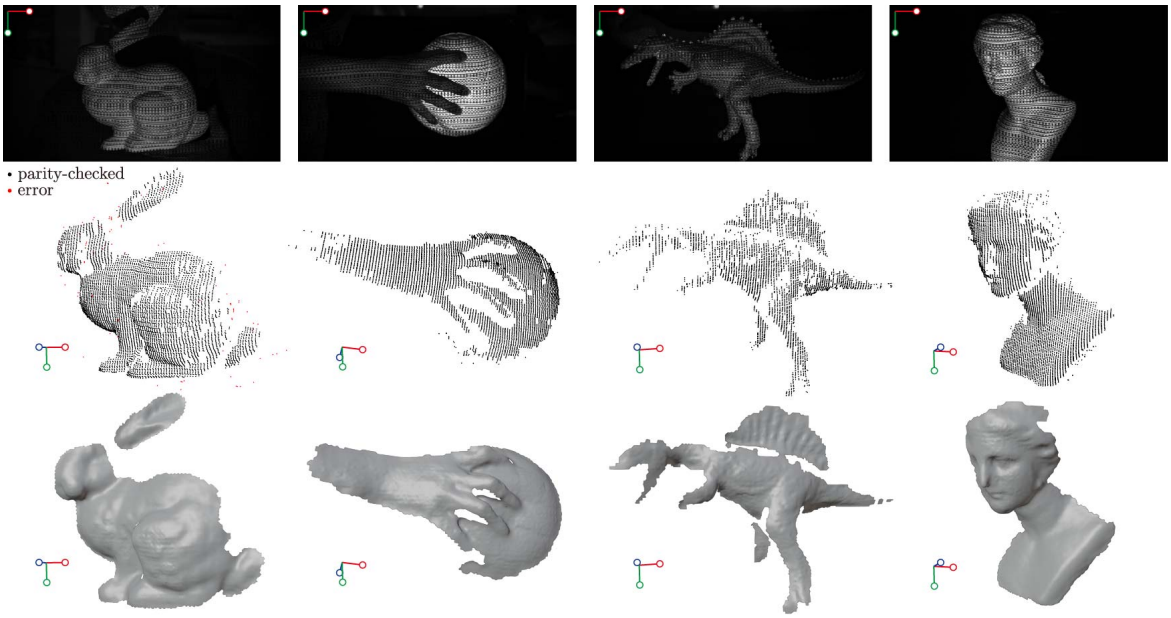


Figure 10. Measurement results of static scenes. From left to right, we measured a Stanford Bunny, volley ball, dinosaur, and the Venus. In the point cloud of the Stanford Bunny, erroneous decoded points are shown in red.

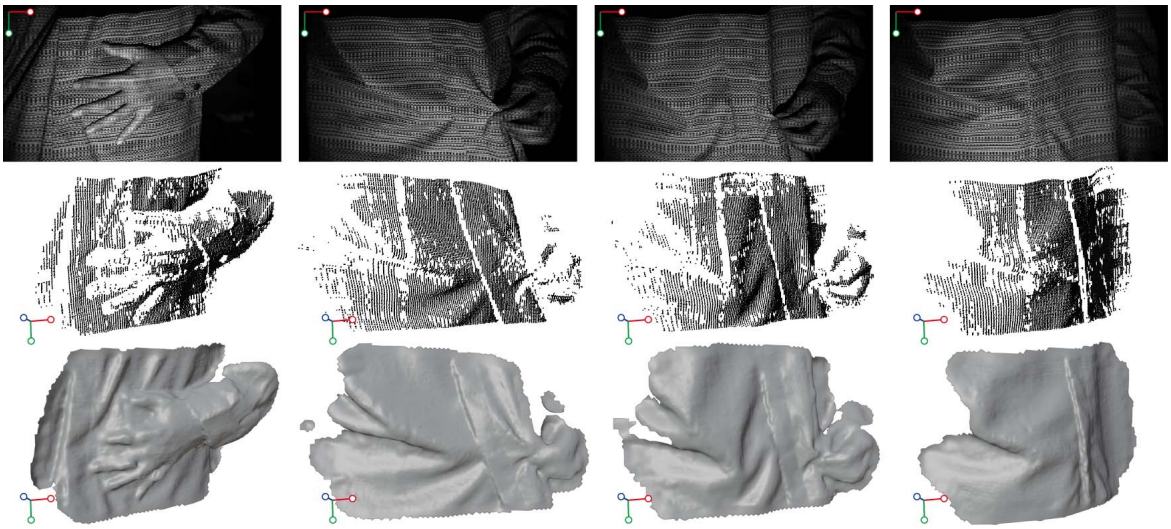


Figure 11. Measurement results of a dynamic scene. The shirt was pinched by the hand.

number of measurement points was 26713, and the density was about $8\ \mathrm{points/cm^2}$ at the center of the image and $6\ \mathrm{points/cm^2}$ at the edges. Although the reduction of the measurement density is inevitable in a spatially encoded pattern, the parallel-bus pattern measured 16.7 times more points than the state-of-the-art high-speed measurement method with a binary monochrome spatially encoded pattern [25]. Besides, the parallel-bus pattern is efficient in data amount per unit time. For example, 8-bit grayscale phase shift patterns yield $1920 \times 1080 \approx 2.0\mathrm{M}$ points per frame by the same devices in our setup, but the parallel-bus pattern can provide $26713 \times 2^8 \approx 6.8\mathrm{M}$ points with the same total exposure time and the exact duration of the LSB bit-plane.

5.3. Measurement accuracy

To evaluate measurement accuracy, we used the result of the 3D-printed Stanford Bunny in Fig. 10 and calculated the error between the model and the measured point cloud. Figure 12 shows the Hausdorff distance from the model data. In the evaluation, the point cloud was aligned to the model data by ICP algorithm [2], and colored based on the dis-

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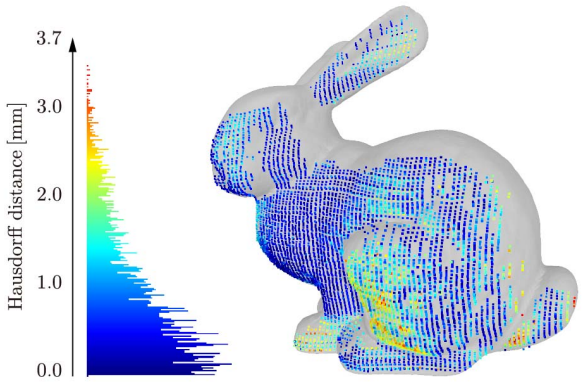


Figure 12. Measurement accuracy. The point cloud is colored by Hausdorff distance from the 3D model.

tance. The Hausdorff distance was 0.838 mm on average and up to 3.71 mm. The figure shows that significant error occurs particularly on highly inclined or discontinuous surfaces. This feature may be familiar to measurement methods using a spatially encoded pattern. In a spatially encoded pattern, decoding involves multiple pixels in a small area. The calculated center position of the area can deviate from the correct position due to a surface inclination. The deviation amount depends on the degree of the inclination and the size of the area, and decoding in the smaller area will cause a minor deviation. Therefore, a smaller unit of a pattern contributes not only to measurement density but also to measurement accuracy. In the proposed method, a compact submatrix realized by the physically stereo-rectified setup and the parallel-bus pattern contributes to improvement in the measurement density and the sub-millimeter measurement accuracy.

5.4. Comparison with related works

Table 1 shows the summarized performance comparison among one-shot 3D sensing methods. Segment pattern (Seg.) [25] is a one-shot binary spatially encoded pattern and the state-of-the-art method which can be fairly compared with the proposed method. Integration method with surface normals (Nor.) [19] uses multiple wavelengths and interpolation, and FreeCam3D (Fre.) [29] uses inference. Besides, the performance of the 1st-generation Kinect [16, 26] in the table is not officially provided.

The latency, resolution and accuracy involve trade-off in structured light methods. As shown in the table, the proposed method has the shortest latency, and the resolution and accuracy are practical as a method without any interpolation and inference.

6. Limitations

The proposed 3D sensing method with the parallel-bus pattern achieves high-speed and low-latency measurement.

Table 1. Performance comparison among one-shot 3D sensing methods. Each column shows latency from image input to point cloud output, frame rate, resolution (Res.), and measurement accuracy (Accu.).

Method	Latency [ms]	Res. [points]	Accu. [mm]
Ours	0.34 (-3000fps)	26.7K	0.84
Seg. [25]	2.1 (-500fps)	1.6K	1.04-2.96
Nor. [19]	7.8 (-400fps)	0.3M	1.61
Fre. [29]	unstated (30fps)	1.1M	3.7
Kinect	267 (-30fps)	76.8K	1.09-10.99

On the other hand, the spatially encoded pattern can not provide pixel-wise measurement in principle. To improve the measurement density, elastic matching methods based on the measured discrete phase and downsizing unit of a pattern by allowing colors and gradation can be potential candidates.

Besides, the spatially encoded pattern can be affected by drastic changes in texture. However, a binary pattern is comparatively robust to the texture like the dinosaur toy in Fig. 10. Moreover, error correction methods used in communication can be used in our measurement scheme and will assist the robustness.

On the other hand, our measurement scheme requires the physically stereo-rectified projector-camera system to achieve high-speed and low-latency decoding. This restriction needs additional alignment of the device components, but even manual adjustment can meet the requirement. Hence, the tolerance will be easily satisfied when the components are mechanically assembled for industrial applications.

7. Conclusion

In this paper, we proposed a high-speed and low-latency 3D sensing system with a structured light pattern named "Parallel-bus pattern." The proposed system assumes a physically stereo-rectified pair of a camera and projector, and the assumption provides an analogy with high-speed communication in a parallel bus interface. To take advantage of the analogy, the parallel-bus pattern is designed with a De Bruijn torus and clock line and efficiently decoded by high-speed sequential memory access and bit operations. As a result, our sensing strategy and the parallel-bus pattern achieved sub-millisecond decoding and sub-millimeter accuracy while improving the measurement density. This technology will help with the high-speed measurement of enormous and dynamic objects and real-time manipulation of unpredictable targets.

Future works include high-speed elastic matching for denser measurement and high-speed simultaneous sensing of shape and texture based on the parallel-bus pattern. Furthermore, we will introduce the proposed system to high-speed robot manipulation and dynamic projection mapping.

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